MAT 3007 OPTIMIZATION
CUHK-SHENZHEN • SDS
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MIDTERM

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Name: _____

Student ID: _____

Question	Points	Score
True/False	15	
The Simplex Method	24	
Duality	20	
Sensitivity Analysis	20	
Modeling	10	
Structured LP	11	
Total:	100	

- Please write down your **name** and **student ID**.
- Please justify your answers except Question 1.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

Question 1 [15 points]: True/False

State whether each of the following statements is *True* or *False*. For each part, only your answer, which should be one of True or False, will be graded. Explanations are not required and will not be read.

(a) [3 points] If an LP in standard form has a unique optimal solution, then the optimal solution must be a basic feasible solution.

Solution: True. By fundamental LP theorem.

(b) [3 points] If the primal problem is unbounded, then the dual problem must be infeasible.

Solution: True. By weak duality.

(c) [3 points] Consider the simplex method applied to an LP in standard form with the matrix A being full row-rank. A variable that has just entered the basis cannot leave in the very next iteration.

Solution: False. *Counterexample:* Consider

$$\begin{aligned} \min \quad & -10x_2 - 2x_3 \\ \text{s.t.} \quad & x_1 + 10x_2 + x_3 = 10 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Iteration 1. The initial basis consists of the single variable x_1. The reduced costs for the nonbasic variables x_2 and x_3 are strictly their objective coefficients:

$$\bar{c}_2 = -10, \qquad \bar{c}_3 = -2.$$

Choosing the most negative reduced cost, x_2 enters the basis. Since there is only one constraint ($m = 1$), the only basic variable available to leave is x_1. Thus, x_1 leaves.

Iteration 2. The new reduced costs are:

$$\bar{c}_1 = 1, \qquad \bar{c}_3 = -1.$$

Because $\bar{c}_3 = -1$ is strictly negative, x_3 enters the basis. Again, because there is only one constraint, the current basic variable x_2 must leave the basis.

Hence, the variable that just entered (x_2) leaves in the very next iteration, proving the statement false.

(d) [3 points] In the two-phase simplex method, the original problem is feasible if and only if the optimal value of the auxiliary problem of phase I is exactly zero.

Solution: True.

(e) [3 points] Consider an LP in standard form with the matrix A being full row-rank. If a non-basic variable has a zero reduced cost in an optimal basic feasible solution, then the LP must have at least two distinct optimal basic feasible solutions.

Solution: False.

Counterexample: Consider

$$\begin{aligned} \min \quad & -x_1 \\ \text{s.t.} \quad & x_1 + x_2 + s_1 = 1 \\ & x_1 + s_2 = 1 \\ & x_1,\, x_2,\, s_1,\, s_2 \geq 0. \end{aligned}$$

It is clear that there is exactly one optimal basic feasible solution (BFS): $(x_1, x_2, s_1, s_2) = (1, 0, 0, 0)$. Take the optimal basis $B = \{x_1, s_1\}$ (so x_1 and s_1 are basic, x_2 and s_2 are non-basic). For the non-basic variable x_2, the reduced cost is exactly 0. Thus, we have an optimal BFS with a non-basic variable (x_2) having zero reduced cost.

Question 2 [24 points]: The Simplex Method

Consider the following LP problem:

$$\begin{aligned} \underset{x_1,x_2,x_3}{\text{maximize}} \quad & 3x_1 + 8x_2 + 9x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 \geq 3, \\ & x_1 + 2x_2 + 3x_3 \leq 6, \\ & 3x_1 + 6x_2 + 5x_3 = 18, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

(a) [4 points] Convert the problem to standard form.

Solution: Introduce a slack variable s_1 for the first constraint and a slack variable s_2 for the second constraint. The standard-form problem is

$$\begin{aligned} \underset{x_1,x_2,x_3,s_1,s_2}{\text{maximize}} \quad & 3x_1 + 8x_2 + 9x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 - s_1 = 3, \quad \text{(1 pts)} \\ & x_1 + 2x_2 + 3x_3 + s_2 = 6, \quad \text{(1 pts)} \\ & 3x_1 + 6x_2 + 5x_3 = 18, \quad \text{(1 pts)} \\ & x_1, x_2, x_3, s_1, s_2 \geq 0. \quad \text{(1 pts)} \end{aligned}$$

(b) [4 points] Write down the Phase I problem and the initial Phase I tableau for the two-phase simplex method.

Solution: Introduce artificial variables y_1, y_2, y_3. The Phase I problem is

$$\begin{aligned} \underset{x_1,x_2,x_3,s_1,s_2,y_1,y_2,y_3}{\text{minimize}} \quad & y_1 + y_2 + y_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 - s_1 + y_1 = 3, \\ & x_1 + 2x_2 + 3x_3 + s_2 + y_2 = 6, \\ & 3x_1 + 6x_2 + 5x_3 + y_3 = 18, \\ & x_1, x_2, x_3, s_1, s_2, y_1, y_2, y_3 \geq 0. \end{aligned}$$

The initial Phase I tableau is

B	-5	-10	-9	1	-1	0	0	0	-27
y_1	1	2	1	-1	0	1	0	0	3
y_2	1	2	3	0	1	0	1	0	6
y_3	3	6	5	0	0	0	0	1	18

(c) [10 points] Apply the smallest index rule for pivoting, and solve the Phase I problem to optimality using simplex tableaus. Please list all intermediate tableaus and the final optimal solution of the Phase I problem.

Solution: Pivot 1: x_1 enters and y_1 leaves.

Among the variables with negative objective-row coefficients, x_1 has the smallest index. The minimum-ratio test gives

$$\frac{3}{1} = 3, \qquad \frac{6}{1} = 6, \qquad \frac{18}{3} = 6.$$

Therefore, y_1 leaves the basis.

After pivoting, the tableau becomes

B	0	0	-4	-4	-1	5	0	0	-12
x_1	1	2	1	-1	0	1	0	0	3
y_2	0	0	2	1	1	-1	1	0	3
y_3	0	0	2	3	0	-3	0	1	9

.

Pivot 2: x_3 enters and y_2 leaves.

The variables x_3, s_1, s_2 have negative objective-row coefficients. By the smallest-index rule, x_3 enters.

The minimum-ratio test gives

$$\frac{3}{1} = 3, \qquad \frac{3}{2} = \frac{3}{2}, \qquad \frac{9}{2} = \frac{9}{2}.$$

Therefore, y_2 leaves the basis.

After pivoting, the tableau becomes

B	0	0	0	-2	1	3	2	0	-6
x_1	1	2	0	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_3	0	0	1	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{3}{2}$
y_3	0	0	0	2	-1	-2	-1	1	6

.

Pivot 3: s_1 enters and x_3 leaves.

The s_1-column is the only column with a negative coefficient in the objective row. Hence, s_1 enters.

The applicable ratios are

$$\frac{\frac{3}{2}}{\frac{1}{2}} = 3 \qquad \text{and} \qquad \frac{6}{2} = 3.$$

There is a tie between the x_3-row and the y_3-row. Since x_3 has the smaller index, x_3 leaves.

After pivoting, the tableau is

B	0	0	4	0	3	1	4	0	0
x_1	1	2	3	0	1	0	1	0	6
s_1	0	0	2	1	1	-1	1	0	3
y_3	0	0	-4	0	-3	0	-3	1	0

.

All reduced costs are nonnegative. Therefore, the optimal value of the Phase I problem is 0.

The corresponding optimal basic feasible solution is

$$x_1 = 6, \qquad s_1 = 3, \qquad y_3 = 0,$$

and

$$x_2 = x_3 = s_2 = y_1 = y_2 = 0.$$

(d) [6 points] Based on the optimal Phase I basis, identify a feasible basic feasible solution and the corresponding basis for the original problem.

Solution:
After deleting the y_3-variable column, the coefficient matrix of the original standard-form problem is

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 & 0 \\ 1 & 2 & 3 & 0 & 1 \\ 3 & 6 & 5 & 0 & 0 \end{bmatrix},$$

where the columns correspond, respectively, to

$$x_1,\ x_2,\ x_3,\ s_1,\ s_2.$$

The non-artificial variables that remain basic after Phase I are x_1 and s_1. Their columns are

$$a_{x_1} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \qquad a_{s_1} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}.$$

The first nonbasic non-artificial variable in the prescribed ordering is x_2. However,

$$a_{x_2} = 2a_{x_1}.$$

x_2 cannot be added to form a basis.
The next candidate is x_3. We have

$$\det \begin{bmatrix} a_{x_1} & a_{s_1} & a_{x_3} \end{bmatrix} = \det \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 3 \\ 3 & 0 & 5 \end{bmatrix} = -4 \neq 0.$$

Therefore, $B = \{x_1, s_1, x_3\}$ is a basis for the original problem.
The feasible basic solution is

$$x_1 = 6, \qquad s_1 = 3, \qquad x_3 = 0,$$

with

$$x_2 = s_2 = 0.$$

Question 3 [20 points]: Duality

Consider the following linear programming problem:

$$\begin{array}{lll} \underset{x_1,x_2,x_3}{\text{minimize}} & x_1 - x_2 + 3x_3 & \\ \text{subject to} & x_1 - x_2 + x_3 & \geq 2, \\ & -x_1 - x_2 & \leq 0, \\ & -2x_1 - x_3 & \geq -2, \\ & x_1 \geq 0,\ x_2 \leq 0,\ x_3 \geq 0. & \end{array}$$

(a) [7 points] Derive the dual problem.

Solution: The dual problem is

$$\begin{aligned} \underset{y_1,y_2,y_3}{\text{maximize}} \quad & 2y_1 - 2y_3 \quad \text{(1 pt)} \\ \text{subject to} \quad & y_1 - y_2 - 2y_3 \le 1 \quad \text{(1 pt)} \\ & -y_1 - y_2 \ge -1 \quad \text{(1 pt)} \\ & y_1 - y_3 \le 3 \quad \text{(1 pt)} \\ & y_1 \ge 0, \quad y_2 \le 0, \quad y_3 \ge 0. \quad \text{(3 pts)} \end{aligned}$$

(b) [7 points] The primal problem has the unique optimal solution

$$\boldsymbol{x}^* = (1, -1, 0)^\top.$$

Using complementary slackness, characterize all optimal solutions of the dual problem. Is the dual optimal solution unique?

Solution: At $\boldsymbol{x}^* = (1, -1, 0)^\top$, the three primal constraints are active.
Since $x_1^* > 0$, the first dual constraint must be active. Since $x_2^* < 0$, the second dual constraint must also be active. Thus, complementary slackness gives

$$\begin{aligned} y_1 - y_2 - 2y_3 &= 1, \quad \text{(1 pt)} \\ -y_1 - y_2 &= -1. \quad \text{(1 pt)} \end{aligned}$$

So

$$y_2 = 1 - y_1.$$

Substituting this into the first equality gives

$$y_1 - (1 - y_1) - 2y_3 = 1,$$

and hence

$$y_1 - y_3 = 1.$$

Let $t = y_3$. We then obtain

$$y_1 = t + 1, \qquad y_2 = -t, \qquad y_3 = t. \quad \text{(2 pts)}$$

The sign restrictions require $t \ge 0$. For every $t \ge 0$,

$$y_1 = t + 1 \ge 0, \qquad y_2 = -t \le 0, \qquad y_3 = t \ge 0.$$

The remaining dual constraint is also satisfied because

$$y_1 - y_3 = (t + 1) - t = 1 \le 3. \quad \text{(1 pt)}$$

Therefore, every vector

$$\boldsymbol{y}(t) = (t + 1, -t, t)^\top, \qquad t \ge 0,$$

is dual feasible and satisfies complementary slackness with $\boldsymbol{x}^*$. Its dual objective value is

$$2y_1 - 2y_3 = 2(t + 1) - 2t = 2.$$

The primal objective value at $\boldsymbol{x}^*$ is

$$x_1^* - x_2^* + 3x_3^* = 1 - (-1) + 3(0) = 2.$$

Hence, every such $\boldsymbol{y}(t)$ is dual optimal by strong duality. (1 pt)
The dual optimal solution is not unique. (1 pt)

(c) [6 points] Let $0 < \varepsilon < 1$. Suppose that the right-hand side of the third primal constraint is changed from -2 to $-2+\varepsilon$, so that the third constraint becomes

$$-2x_1 - x_3 \geq -2 + \varepsilon.$$

Derive the modified dual problem and show that it is unbounded. What does this imply about the feasibility of modified primal problem?

Solution: The modified dual is

$$\begin{aligned} \underset{y_1,y_2,y_3}{\text{maximize}} \quad & 2y_1 + (-2+\varepsilon)y_3 \quad \text{(1 pt)} \\ \text{subject to} \quad & y_1 - y_2 - 2y_3 \leq 1, \\ & -y_1 - y_2 \geq -1, \\ & y_1 - y_3 \leq 3, \\ & y_1 \geq 0, \quad y_2 \leq 0, \quad y_3 \geq 0. \end{aligned}$$

Consider

$$\boldsymbol{y}(t) = (t+1, -t, t)^\top, \qquad t \geq 0.$$

It is clear that $\boldsymbol{y}(t)$ is feasible for the modified dual problem for every $t \geq 0$. (2 pts)
The modified dual objective value is

$$\begin{aligned} 2y_1 + (-2+\varepsilon)y_3 &= 2(t+1) + (-2+\varepsilon)t \\ &= 2 + \varepsilon t. \end{aligned}$$

Since $\varepsilon > 0$,

$$2 + \varepsilon t \longrightarrow +\infty \qquad \text{as } t \longrightarrow +\infty.$$

Therefore, the modified dual problem is unbounded. (2 pts)
By weak duality, if the modified dual problem is unbounded, then the modified primal problem must be infeasible. (1 pt)

Question 4 [20 points]: Sensitivity Analysis

Consider the following linear programming problem:

$$\begin{aligned} \underset{x_1,x_2,x_3}{\text{minimize}} \quad & -5x_1 - 4x_2 - 6x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 \leq 8, \\ & 2x_1 + x_2 + x_3 \leq 9, \\ & x_1 + x_2 + 2x_3 \leq 7, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

After converting the problem into standard form (by adding slack variables s_1, s_2, s_3) and solving it, we obtain the final optimal simplex tableau:

B	0	0	0	$\frac{1}{4}$	$\frac{5}{4}$	$\frac{9}{4}$	29
1	1	0	0	$-\frac{1}{4}$	$\frac{3}{4}$	$-\frac{1}{4}$	3
2	0	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	$-\frac{1}{4}$	2
3	0	0	1	$-\frac{1}{4}$	$-\frac{1}{4}$	$\frac{3}{4}$	1

From the optimal simplex tableau, answer the following questions.

(a) [3 points] What are the optimal solution and the optimal objective value of the original linear program?

Solution: From the final tableau, the basic variables are x_1, x_2, x_3, with

$$x_1^* = 3, \qquad x_2^* = 2, \qquad x_3^* = 1.$$

Hence, the optimal solution is

$$\boldsymbol{x}^* = (3, 2, 1)^\top. \qquad \text{(2pts)}$$

The optimal value is -29. (1pt)

(b) [6 points] In what range can we change the right-hand side $b_1 = 8$ of the first constraint so that the current optimal basis of the standard linear program remains optimal?

Solution: Let the new right-hand side be

$$b_1 = 8 + \lambda.$$

The current basis is $B = \{1, 2, 3\}$ and the current basic solution is $\boldsymbol{x}_B^* = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$.

Because s_1, s_2, s_3 are the slack variables, the columns under s_1, s_2, s_3 in the final tableau give

$$A_B^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{bmatrix}.$$

Therefore,

$$A_B^{-1}\boldsymbol{e}_1 = \begin{bmatrix} -\frac{1}{4} \\ \frac{3}{4} \\ -\frac{1}{4} \end{bmatrix}. \qquad \text{(2pts)}$$

Changing only the right-hand side does not change the reduced costs. Therefore, the current basis remains optimal if and only if the new basic solution remains feasible:

$$\boldsymbol{x}_B^* + \lambda A_B^{-1}\boldsymbol{e}_1 \geq 0.$$

That is,

$$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} -\frac{1}{4} \\ \frac{3}{4} \\ -\frac{1}{4} \end{bmatrix} \geq 0.$$

Equivalently,

$$3 - \frac{\lambda}{4} \geq 0, \qquad 2 + \frac{3\lambda}{4} \geq 0, \qquad 1 - \frac{\lambda}{4} \geq 0. \qquad \text{(2pts)}$$

These inequalities imply

$$-\frac{8}{3} \leq \lambda \leq 4.$$

Since $b_1 = 8 + \lambda$, the allowable range is

$$\boxed{\frac{16}{3} \leq b_1 \leq 12.} \qquad \text{(2pts)}$$

(c) [6 points] In what range can we change the objective coefficient $c_2 = -4$ of x_2 so that the current optimal basis of the standard linear program remains optimal?

Solution: Let the new objective coefficient of x_2 be

$$c_2 = -4 + \lambda.$$

Thus, the basic cost vector becomes

$$\boldsymbol{c}_B(\lambda) = \boldsymbol{c}_B + \lambda \boldsymbol{e}_2. \qquad \text{(1pt)}$$

The nonbasic variables are s_1, s_2, s_3, and their current reduced costs are

$$\overline{\boldsymbol{c}}_N^\top = \begin{bmatrix} \frac{1}{4} & \frac{5}{4} & \frac{9}{4} \end{bmatrix}.$$

Since $A_N = I_3$, the new reduced-cost vector is

$$\begin{aligned} \overline{\boldsymbol{c}}_N(\lambda)^\top &= \boldsymbol{c}_N^\top - \boldsymbol{c}_B(\lambda)^\top A_B^{-1} A_N \\ &= \overline{\boldsymbol{c}}_N^\top - \lambda \boldsymbol{e}_2^\top A_B^{-1} A_N. \end{aligned}$$

The second row of A_B^{-1} is

$$\boldsymbol{e}_2^\top A_B^{-1} A_N = \begin{bmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}.$$

Therefore,

$$\overline{\boldsymbol{c}}_N(\lambda)^\top = \begin{bmatrix} \frac{1}{4} - \frac{3\lambda}{4} & \frac{5}{4} + \frac{\lambda}{4} & \frac{9}{4} + \frac{\lambda}{4} \end{bmatrix}. \qquad \text{(2pts)}$$

For the current basis to remain optimal in a minimization problem, all reduced costs must remain nonnegative:

$$\frac{1}{4} - \frac{3\lambda}{4} \geq 0, \qquad \frac{5}{4} + \frac{\lambda}{4} \geq 0, \qquad \frac{9}{4} + \frac{\lambda}{4} \geq 0. \qquad \text{(2pts)}$$

These conditions give

$$-5 \leq \lambda \leq \frac{1}{3}.$$

Since $c_2 = -4 + \lambda$, the allowable range of the objective coefficient is

$$\boxed{-9 \leq c_2 \leq -\frac{11}{3}.} \qquad \text{(1pt)}$$

(d) [5 points] Suppose a new decision variable x_4 is added to the linear program. This new variable has an objective coefficient

$$c_4 = -4$$

and constraint coefficients

$$A_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Determine whether the current optimal basis remains optimal. If not, identify which variable will enter the basis and which will leave.

Solution: The reduced cost of the new variable is

$$\bar{c}_4 = c_4 - A_4^\top A_B^{-\top} \boldsymbol{c}_B = -4 - \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{4} \\ -\frac{5}{4} \\ -\frac{9}{4} \end{bmatrix} = -4 + \frac{15}{4} = -\frac{1}{4}. \quad \text{(2pts)}$$

Since

$$\bar{c}_4 = -\frac{1}{4} < 0,$$

the current basis is not optimal for the expanded minimization problem. Therefore, x_4 is eligible to enter the basis. Since it is the only variable with a negative reduced cost,

$$\boxed{x_4 \text{ enters the basis.}} \quad \text{(1pt)}$$

The basic direction of x_4 is

$$A_B^{-1} A_4 = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{bmatrix}. \quad \text{(1pt)}$$

If the simplex method is continued, the minimum-ratio test gives

$$\min\left\{\frac{3}{1/4}, \frac{2}{1/4}, \frac{1}{1/4}\right\} = 4,$$

so x_3 leaves the basis. (1pt)

Question 5 [10 points]: Modeling

A fund manager must invest all available capital among six investment options. The allocations are expressed as fractions of the total fund (for example, 0.25 means that 25% of the fund is invested in that specific option). The expected returns, the manager's current allocations, and maximum allowable allocations are shown below:

Investment option i	1	2	3	4	5	6
Expected return	6%	8%	10%	7%	9%	7.5%
Current allocation	40%	30%	20%	10%	0%	0%
Maximum allocation	50%	45%	50%	40%	35%	45%

The fund manager wants to maximize the expected return of the new portfolio while satisfying the following portfolio requirements.

- All available capital must be invested.
- The manager can select at most three investment options.
- If an investment option is selected, at least 10% of the fund must be invested in it. Its allocation must also not exceed the corresponding maximum allocation in the table.
- Options 2 and 5 cannot both be selected.
- The total absolute change from the current portfolio to the new one cannot exceed 40%.

(a) [4 points] Define all decision variables needed to formulate the problem as a mixed-integer linear program. Clearly state what each variable represents and outline its domain.
Hint: A mixed-integer linear program is an optimization problem in which the objective function and all constraints are linear, while some decision variables may be integer.

Solution: For each $i = 1, \ldots, 6$, define the following variables:

- Let x_i denote the fraction of the fund invested in investment option i, where
$$x_i \geq 0. \quad \text{(1pt)}$$
- Let
$$z_i = \begin{cases} 1, & \text{if investment option } i \text{ is selected,} \\ 0, & \text{otherwise,} \end{cases} \qquad z_i \in \{0, 1\}. \quad \text{(2pts)}$$
- Let $d_i \geq 0$ be an auxiliary variable used to represent the absolute change in the allocation of option i, namely
$$d_i \geq \left| x_i - x_i^{(0)} \right|. \quad \text{(1pt)}$$
The absolute value will be linearized in part (b).

(b) [6 points] Formulate a mixed-integer linear program that maximizes the expected return of the portfolio while satisfying all the requirements above.

Solution: Let
$$\left(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, x_4^{(0)}, x_5^{(0)}, x_6^{(0)}\right) = (0.4, 0.3, 0.2, 0.1, 0, 0),$$
and let
$$(u_1, u_2, u_3, u_4, u_5, u_6) = (0.50, 0.45, 0.50, 0.40, 0.35, 0.45)$$
denote the maximum allocation levels. The problem can be formulated as follows:

$$\begin{aligned}
\underset{\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{d}}{\text{maximize}} \quad & 0.06x_1 + 0.08x_2 + 0.10x_3 + 0.07x_4 + 0.09x_5 + 0.075x_6 \quad \text{(1pt)} \\
\text{subject to} \quad & \sum_{i=1}^{6} x_i = 1 \quad \text{(1pt)} \\
& 0.10z_i \leq x_i \leq u_i z_i, \qquad i = 1, \ldots, 6, \quad \text{(1pts)} \\
& \sum_{i=1}^{6} z_i \leq 3 \quad \text{(1pt)} \\
& z_2 + z_5 \leq 1 \quad \text{(1pt)} \\
& d_i \geq x_i - x_i^{(0)}, \qquad i = 1, \ldots, 6, \\
& d_i \geq x_i^{(0)} - x_i, \qquad i = 1, \ldots, 6, \\
& \sum_{i=1}^{6} d_i \leq 0.4 \quad \text{(1pts)} \\
& x_i \geq 0, \quad d_i \geq 0, \qquad i = 1, \ldots, 6, \\
& z_i \in \{0, 1\}, \qquad i = 1, \ldots, 6.
\end{aligned}$$

Note: Other equivalent linearizations of the absolute values are also acceptable. However, writing only $\sum_{i=1}^{6} |x_i - x_i^{(0)}| \leq 0.4$ does not provide a mixed-integer linear formulation.

Question 6 [11 points]: Structured LP

Consider the following linear program:

$$\begin{aligned} \underset{x_1,\cdots,x_n}{\text{minimize}} \quad & \sum_{i=1}^{n} x_i \\ \text{subject to} \quad & \sum_{i=1}^{n} i x_i \geq n \\ & \sum_{i=1}^{n} (n+1-i) x_i \geq n \\ & x_1, \cdots, x_n \geq 0. \end{aligned}$$

(a) [3 points] Determine whether the feasible region is nonempty and bounded. Justify your answer.

Solution: **Feasibility:** Consider the point

$$x_1 = 1, \quad x_n = 1, \quad x_i = 0 \ (i \neq 1, n). \quad \text{(1pts)}$$

Then we have

$$\sum_{i=1}^{n} i x_i = 1 + n \geq n, \qquad \sum_{i=1}^{n} (n+1-i) x_i = n + 1 \geq n.$$

Hence the feasible region is nonempty.

Unboundedness of the feasible region: The feasible region is unbounded. To see this, consider the solution

$$x_1 = t, \quad x_n = t, \quad x_i = 0 \ (i \neq 1, n), \quad t \geq 0. \quad \text{(1pts)}$$

Then

$$\sum_{i=1}^{n} i x_i = t + nt = (n+1)t \geq n, \qquad \sum_{i=1}^{n} (n+1-i) x_i = nt + t = (n+1)t \geq n,$$

for all sufficiently large t. Thus the constraints remain satisfied as $t \to \infty$, showing that the feasible region is unbounded. (1pts) ■

(b) [3 points] Prove that every extreme point of the feasible region has at most two nonzero components.

Solution: We first convert the problem into standard form by introducing slack variables $s_1, s_2 \geq 0$:

$$\sum_{i=1}^{n} i x_i - s_1 = n, \qquad \sum_{i=1}^{n} (n+1-i) x_i - s_2 = n, \qquad x, s_1, s_2 \geq 0. \quad \text{(1pts)}$$

Let the feasible region be

$$P = \{x \in \mathbb{R}^{n+2} : Ax = b, \ x \geq 0\}.$$

In standard form, there are 2 equality constraints. Hence every BFS is obtained by selecting exactly 2 basic variables and setting the rest to zero.
Therefore, any BFS has at most 2 nonzero variables among

$$\{x_1, \ldots, x_n, s_1, s_2\}. \quad \text{(1pts)}$$

A point $\boldsymbol{x}$ an extreme point of the original inequality-form polyhedron if and only if $\boldsymbol{x}, s_1, s_2$ is an extreme point (i.e., a BFS) of the standard-form polyhedron. Thus, at most two of the x_i can be positive. (1pts) ■

(c) [5 points] Prove that if the optimal solution is unique, then

$$x_i^* = x_{n+1-i}^*.$$

Solution: Observe that the two constraints have perfectly symmetric structure:
- coefficients in the first constraint: $1, 2, \ldots, n$ - coefficients in the second constraint: $n, n-1, \ldots, 1$
Thus the system remains unchanged if we reverse the order of variables. (2pts)

Now take any feasible solution x. If we reverse its coordinates (i.e., swap x_1 with x_n, x_2 with x_{n-1}, etc.), the resulting vector is also feasible and has the same objective value $\sum x_i$. (2pts)

Hence, every feasible solution has a symmetric counterpart with identical cost.
If the optimal solution is unique, then it must coincide with its symmetric counterpart. Therefore,

$$x_i^* = x_{n+1-i}^*, \quad \forall i. \quad \text{(1pts)}$$

■